

FREEBITES

Catering Leftovers | BU Dining Sustainability

PROJECT GOAL

The Problem: Massachusetts generates over 1 million tons of food waste each year, while food insecurity continues to rise. At BU, leftover catered food is often thrown away because staff lack a fast, accessible, and real-time way to alert students.

The Solution: Create a native mobile app that lets BU Dining staff instantly post leftover events and automatically notify students. This provides a quick, reliable, and student-friendly system that reduces waste and improves student access.

OUR CLIENTS



Lawrence Alabaster
Director of Catering
@ BU Dining



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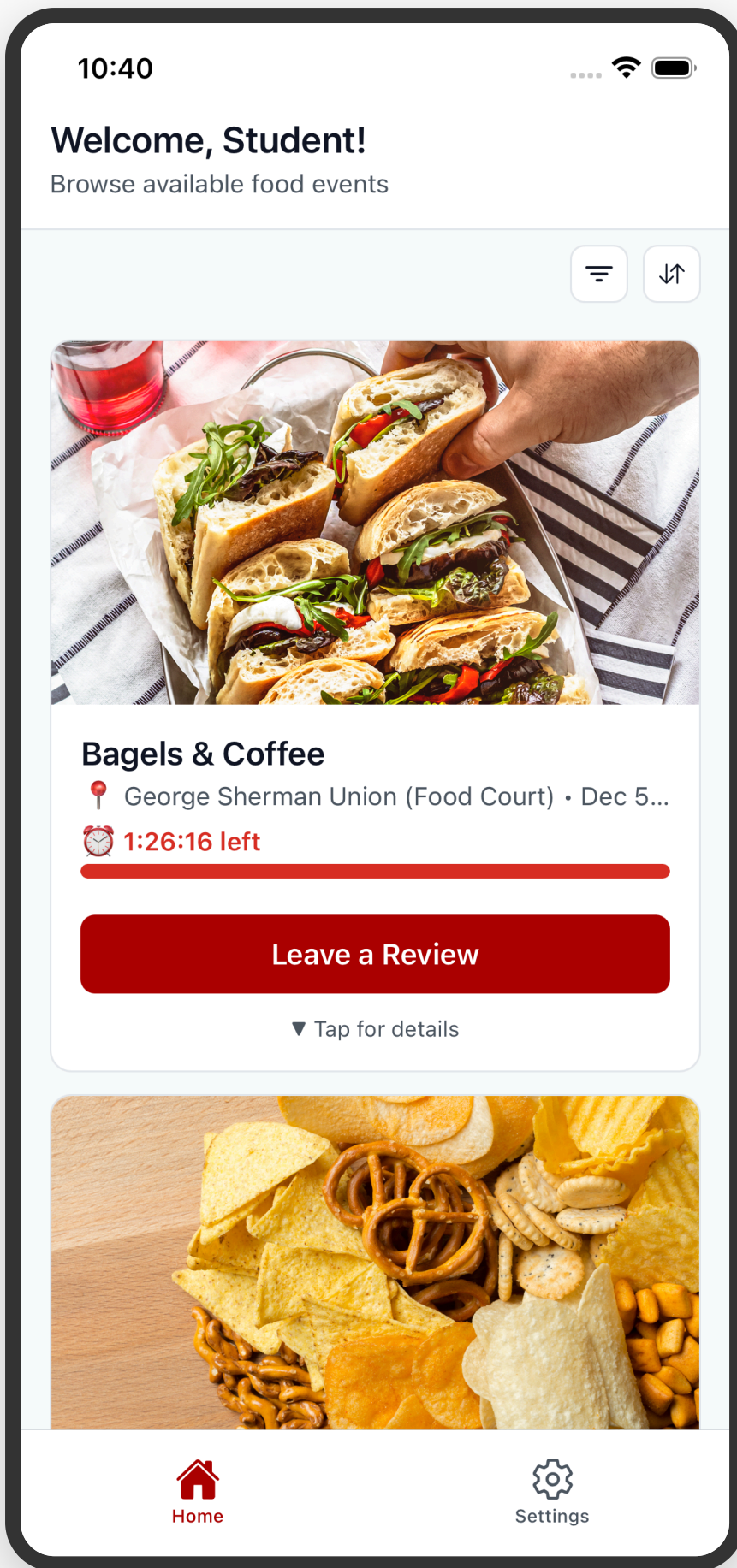
MENTORS

Professor: Nagendra Mishr, Ricardo Villamil

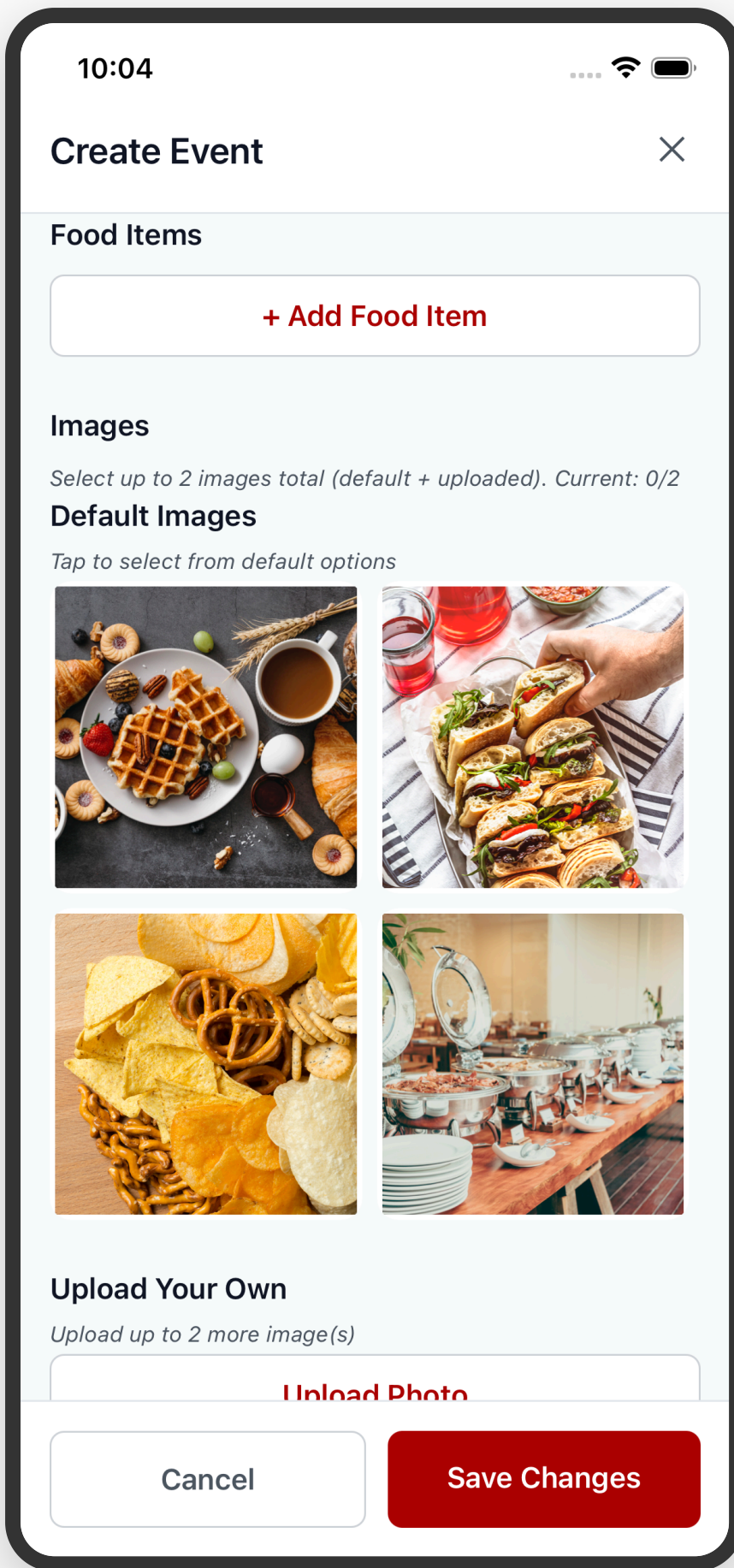
PM & TPM: Kwabena Ampomah, Arnav Singh

TECH STACK

- **Frontend:** React Native / Expo
- **Backend:** Firebase (Firestore)
- **Auth:** Clerk



Student View



Admin View

Key Features

- ✓ Real-time Push Notifications
- ✓ Admin Verification Tools
- ✓ Live Countdown Timers
- ✓ Event Feedback System

FINAL DELIVERABLES

- Rebuilt and migrated the leftover alert system from the legacy web app to a mobile React Native app
- Designed a streamlined "Create Event" flow for fast staff posting
- Implemented RBAC to verify authorized dining staff
- Built real-time push notification delivery
- Developed Firestore-backed event management backend
- Added new mobile-only features such as countdown timers

REFLECTIONS

Building a production-ready system pushed us beyond UI into deeper backend work. Our biggest challenge was setting up RBAC, implementing push notifications, and making sure real-time updates worked smoothly for staff. Client feedback drove several iterations of the event flow and notifications, showing us how operational constraints shape design and how small usability changes can greatly improve staff efficiency.

Next Steps: Deploy to App Store and Google Play.

FUTURE IMPACT

This system lays the foundation for a campus-wide leftover recovery network. With mobile-first workflows, verified staff access, and real-time student reach, FreeBites can scale across BU Dining to significantly reduce food waste.